

Nick Coffman

✉ nicholastimothycoffman@gmail.com
🌐 github.com/nicholastimothycoffman

☎ (217) 971-6997
🌐 linkedin.com/in/nicholastimothycoffman

🌐 nicholastimothycoffman.github.io

Technical Skills

Languages:

Proficient: Python, Java, C++,
Familiar: JavaScript, C, HTML, CSS

Distributed Event Streaming Platforms:

Familiar: Apache Kafka

Quantum Computing:

Proficient: Qiskit

Databases:

Familiar: PostgreSQL, Redis

APIs and Middleware:

Familiar: RESTful Services

Deployment:

Familiar: Docker

Frameworks and Libraries:

Proficient: Flask, Django
Familiar: PyTorch, Matplotlib, TensorFlow, Scikit-learn, Pandas, NumPy

Version Control:

Proficient: GitHub

Document Preparation Systems:

Proficient: LaTeX

Operating Systems:

Proficient: MacOS
Familiar: Linux, Windows

Notebooks:

Proficient: Jupyter

Relevant Experience & Projects

• NeuroSim

Dec '24 - Present

- i) Developing a full-stack web platform to simulate brain region interactions under various conditions, integrating React for frontend and Flask for backend functionalities.
- ii) Implementing machine learning and quantum computing techniques using TensorFlow, PyTorch, Qiskit, and Amazon Braket to optimize and predict brain activity simulations.
- iii) Utilizing 3D brain visualization tools like Three.js and D3.js to provide interactive and detailed insights into brain functionalities and simulation results.
- iv) Designing RESTful APIs to handle simulation scenarios, user input configurations, and results retrieval, enhancing user interaction and data management.
- v) Leveraging AWS SageMaker and Google AI Platform for scalable machine learning model hosting, and employing AWS Lambda for efficient serverless compute operations.
- vi) Incorporating principles of neuroscience such as neuroplasticity and Hebbian learning to ensure the accuracy of simulations, particularly for scenarios affecting cognitive functions like memory and attention.

• Quantum Visualizer

Feb '24 - Apr '24

- i) Developed the "Quantum Visualizer", an educational tool designed to demystify complex quantum computing concepts such as superposition, amplitude amplification, and interference through interactive simulations.
- ii) Utilized IBM's Qiskit framework within a Jupyter Notebook environment to create dynamic visualizations of quantum states and operations, enhancing the intuitive understanding of quantum mechanics.
- iii) Demonstrated key quantum algorithms, including Grover's Algorithm, with visual aids to explain the quantum search process and its efficiency over classical methods.
- iv) Implemented various quantum circuits using Qiskit to show real-time changes in qubit states on the Bloch sphere, providing a hands-on learning experience.
- v) Conducted comprehensive simulations to illustrate the practical application of quantum gates, superposition, and interference in solving complex computational problems.

• Quantum Key Distribution Protocol Demonstration

Feb '24 - Apr '24

- i) Developed and demonstrated the BB84 protocol using IBM's Qiskit to simulate secure cryptographic key exchange using quantum mechanics, enhancing information security against quantum-enabled threats.
- ii) Implemented quantum circuits to simulate the transmission of encrypted keys between parties, ensuring the ability to detect and mitigate eavesdropping attempts using quantum superposition and measurement techniques.
- iii) Utilized AerSimulator for dynamic simulation of quantum behavior on classical hardware, verifying the protocol's ability to secure communication channels even in the presence of potential cyber threats.
- iv) Conducted extensive simulations to illustrate both uncompromised and eavesdropped scenarios, showcasing the protocol's robustness by detecting anomalies introduced by interceptors.
- v) Demonstrated the practical application of quantum cryptography principles such as superposition and entanglement to establish a new standard in secure communication.

Foreign Languages

French: *Fluent*
Spanish: *Intermediate*

German: *Basic Knowledge*
Italian: *Basic Knowledge*

Danish: *Basic Knowledge*
Latin: *Basic Knowledge*

Education

• Texas State University, San Marcos - B.A., Computer Science & French

July '24

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Software Engineering, Computer Architecture, Computer System Security, Network Science, Operating Systems